

Polish Math Olympiad 2007 Round 1 — P6/12

Jonathan Kasongo

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Find all polynomials $W(x)$ such that $W(x^2)W(x^3) = W(x)^5$ holds for every real number x .

Solution

Let the roots of W be $|r_1| \leq |r_2| \leq \dots \leq |r_n|$ where $(r_n) \in \mathbb{C}$. The equation looks like so,

$$(x^2 - r_1)(x^3 - r_1) \cdots (x^2 - r_n)(x^3 - r_n) = (x - r_1)^5 \cdots (x - r_n)^5$$

and so the set of roots of both sides of this equation must be the same.

Claim 1: If r_i is a root of W , then $\pm\sqrt{r_i}$ and $\sqrt[3]{r_i}$ are roots of W as well.

Proof: On the LHS of the above equation we will find the factors $(x^2 - r_i)(x^3 - r_i)$ which have solutions $\pm\sqrt{r_i}$ and $\sqrt[3]{r_i}$. $\square$

Claim 2: $r_1 = r_2 = \dots = r_n = 0$.

Proof: Suppose, for the sake of contradiction, that there exists an integer k such that $r_k \neq 0$.

In fact let k^* be the smallest such integer for which $r_{k^*} \neq 0$. Now since $\sqrt{r_{k^*}} \neq 0$ is a root of the LHS, it is also a root of the RHS. But since $\sqrt{r_{k^*}} \leq r_{k^*}$ and k^* is the smallest integer for which $r_{k^*} \neq 0$, it must be the case that $\sqrt{r_{k^*}} = r_{k^*}$. Since it cannot be 0, we have $r_{k^*} = 1$. Now consider the following, all due to claim 1:

$$\begin{aligned} 1 \text{ is a root of } W &\implies -\sqrt{1} = -1 \text{ is a root of } W &\implies \\ \sqrt{-1} = i \text{ is a root of } W &\implies \sqrt{i} = e^{i\pi/2} \text{ is a root of } W &\implies \\ \sqrt{e^{i\pi/2}} = e^{i\pi/4} \text{ is a root of } W &\implies \sqrt{e^{i\pi/4}} = e^{i\pi/8} \text{ is a root of } W &\implies \\ \dots &\implies \dots &\implies \end{aligned}$$

and thus $e^{i\pi/2^n}$ is a root of W for any positive integer n . However since a polynomial can't have infinite roots, we have reached a contradiction. $\square$

So since $r_1 = r_2 = \dots = r_n = 0$, it must be the case that $W(x) = x^n$ for positive integer n , and this works since $x^{2n} \cdot x^{3n} = (x^n)^5$. $\blacksquare$